SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2_AT1 — CHART CONSTRUCTION TEST

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO2 – Construction (BL3)	**Assessment:**	Assessment Tool 1 – Chart Construction
Weightage:	20% of CIA	**Total Marks:**	25 (5 Questions × 5 Marks)
CO2 Statement:	*Apply appropriate chart construction techniques — correct chart type, axes, scales, and calculations — to accurately represent given data.(BL3)*		
Student Name:	PRIYANKA.G	**Reg. No.:**	192524124
Section / Batch:		**Date:**	30.07.2026

Code of Conduct

I Priyanka. G / 192524124 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ____________________________

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 75 minutes.
- Graph paper or a ruler is recommended for accurate, to-scale drawing.
- Show all calculations (e.g., percentages, angles, bin ranges, five-number summaries) as part of your answer.
- Every chart must include a title, correctly labeled axes, and a legend where applicable.

Annexure A – CHART CONSTRUCTION

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Problem 1 — Bar Chart — Basic Construction (5 Marks)

Given Data:

Month	Sales (Units)
Jan	120
Feb	150
Mar	90
Apr	200
May	175

Task: ***Construct a bar chart to represent monthly sales. Label both axes, include a chart title, and keep the months in chronological order.***

Construction Plan (Key Elements):

Chart Type	Vertical (column) bar chart
X-Axis	Month — nominal/ordinal category, ordered chronologically (Jan → May)
Y-Axis	Sales (Units) — numeric, quantitative
Axis Scale	0 to 200, in intervals of 50 (5 gridlines)
Title	"Monthly Sales Analysis"

Construction Working / Explanation:

The value axis (Y) is scaled from 0 to 200 in equal steps of 50 units (0, 50, 100, 150, 200). This range was chosen because the maximum data value is 200 (April), so scaling exactly to that maximum — using a round, easy-to-read interval — avoids wasted space while still allowing every bar height to be read accurately against the gridlines.

- Jan = 120 units → bar reaches just under the 150 gridline.
- Feb = 150 units → bar reaches exactly the 150 gridline.
- Mar = 90 units → shortest bar, just under the 100 gridline.
- Apr = 200 units → tallest bar, reaching the top gridline.
- May = 175 units → second-tallest bar, midway between 150 and 200.

Each month is plotted as one bar of equal width with uniform spacing, positioned along the category (X) axis strictly in calendar order (Jan, Feb, Mar, Apr, May) rather than alphabetical or magnitude order, since the task specifically requires chronological sequencing so that the trend in sales over time can be read naturally from left to right.

Constructed Chart:

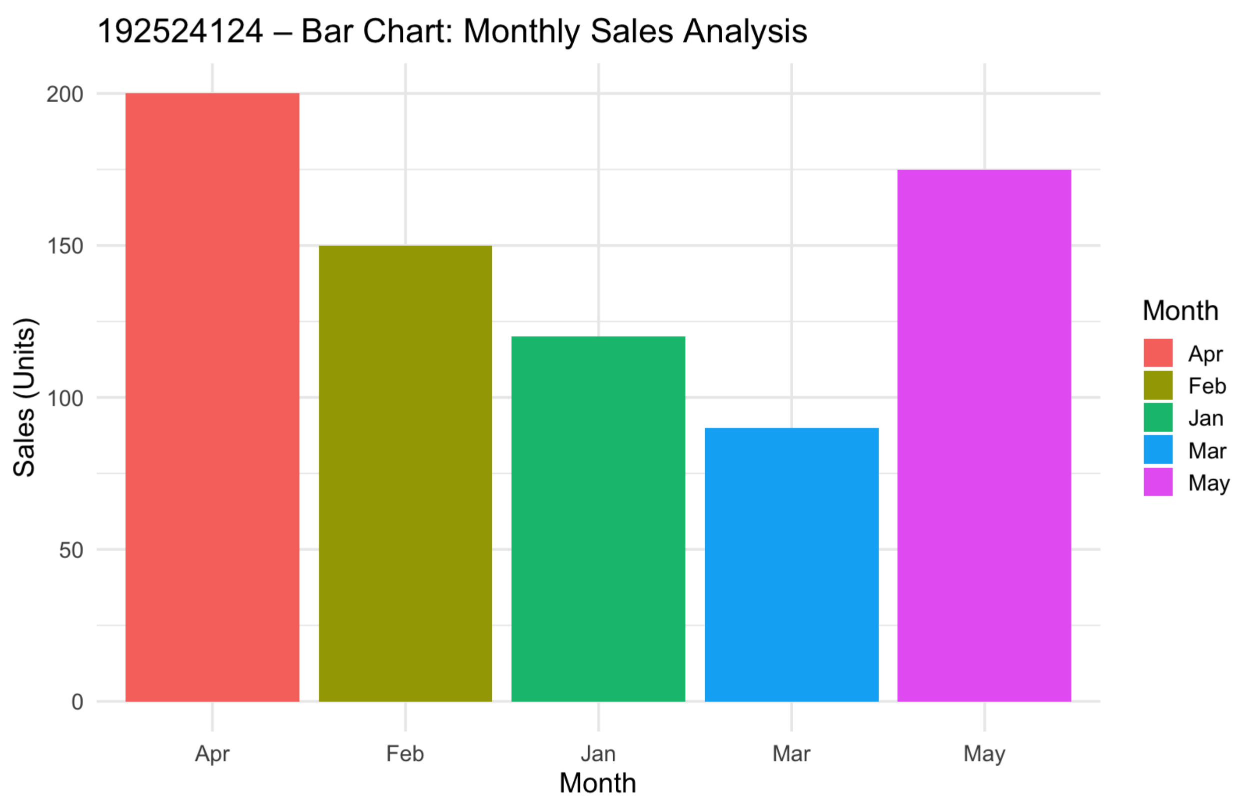

Figure 1: Bar Chart — Monthly Sales Analysis

Problem 2 — Horizontal Bar Chart — Ranked (5 Marks)

Given Data:

Source	Visits
Organic Search	4500
Direct	2200
Social Media	1800
Referral	900
Email	600

Task: ***Construct a horizontal bar chart ranking the traffic sources from highest to lowest. Choose and label appropriate scale intervals on the value axis.***

Construction Plan (Key Elements):

Chart Type	Horizontal bar chart
Y-Axis	Traffic Source — nominal category, ranked by value (highest at top)
X-Axis	Visits — numeric, quantitative
Axis Scale	0 to 4,500, in intervals of 1,000 (rounded up from the 4,500 maximum for clean gridlines)
Title	"Website Traffic by Source"

Construction Working / Explanation:

Since the maximum value in the dataset is 4,500 (Organic Search), the value axis is scaled from 0 up to 5,000 using round intervals of 1,000 (0, 1000, 2000, 3000, 4000, 5000), giving evenly spaced, easy-to-read gridlines that comfortably accommodate the largest bar.

The five categories are sorted in descending order of visits before plotting — Organic Search (4,500), Direct (2,200), Social Media (1,800), Referral (900), Email (600) — so that the chart itself communicates the ranking visually: the largest bar sits at the top and each subsequent bar is shorter, letting a viewer read the relative importance of each channel at a glance without needing to compare exact values.

Horizontal orientation was chosen over a vertical layout because it gives the category labels (traffic source names) more horizontal room to display fully without truncation or rotation, which is a common best practice when category names are long or numerous.

Constructed Chart:

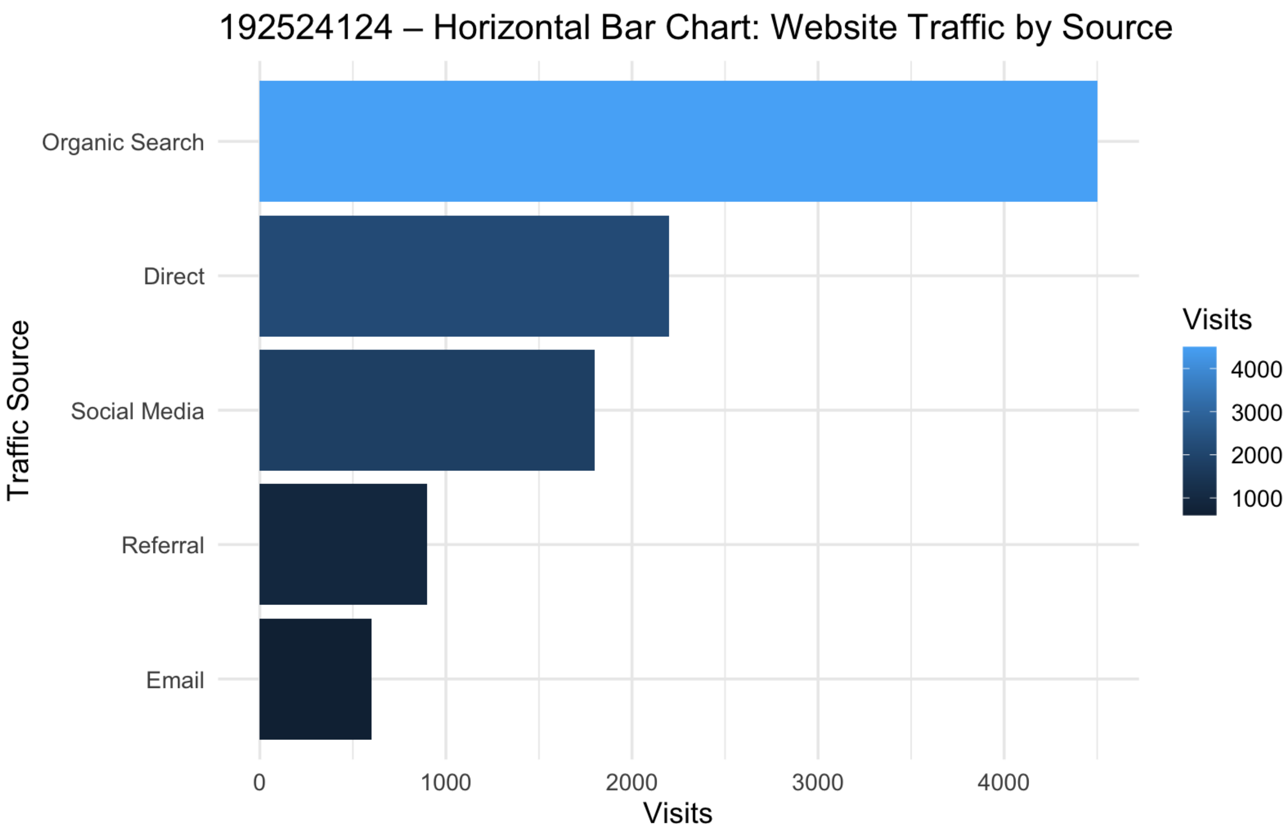

Figure 2: Horizontal Bar Chart — Website Traffic

Problem 3 — Grouped (Clustered) Bar Chart (5 Marks)

Given Data:

Quarter	Q1	Q2	Q3	Q4
Product A ($'000)	200	250	300	280
Product B ($'000)	180	220	260	310

Task: ***Construct a grouped bar chart comparing Product A and Product B across all four quarters. Include a legend distinguishing the two products.***

Construction Plan (Key Elements):

Chart Type	Grouped (clustered) bar chart
X-Axis	Quarter — ordinal category (Q1 → Q4)
Y-Axis	Revenue ($'000) — numeric, quantitative
Axis Scale	0 to 300+, in intervals of 100 (rounded up from the 310 maximum)
Aesthetic Mapping / Legend	Colour hue = Product (A = red, B = teal)

Construction Working / Explanation:

The value axis is scaled from 0 to just above 300 (the highest value, 310, occurs for Product B in Q4), using round intervals of 100 so both series can be compared on a common, easily-read scale.

For each quarter, two bars of equal width are placed directly next to each other (clustered) rather than stacked, so that the heights of Product A and Product B can be compared side by side within the same quarter, as well as across quarters:

- Q1: A = 200, B = 180 — A is taller.
- Q2: A = 250, B = 220 — A is taller.
- Q3: A = 300, B = 260 — A is taller, and this is A's peak quarter.
- Q4: A = 280, B = 310 — B overtakes A for the first time, becoming the taller bar.

A consistent colour is used for each product across all four quarters (rather than a new colour per quarter), and this mapping is explained in a legend, so the viewer can track each product's trend across the full year using colour alone.

Constructed Chart:

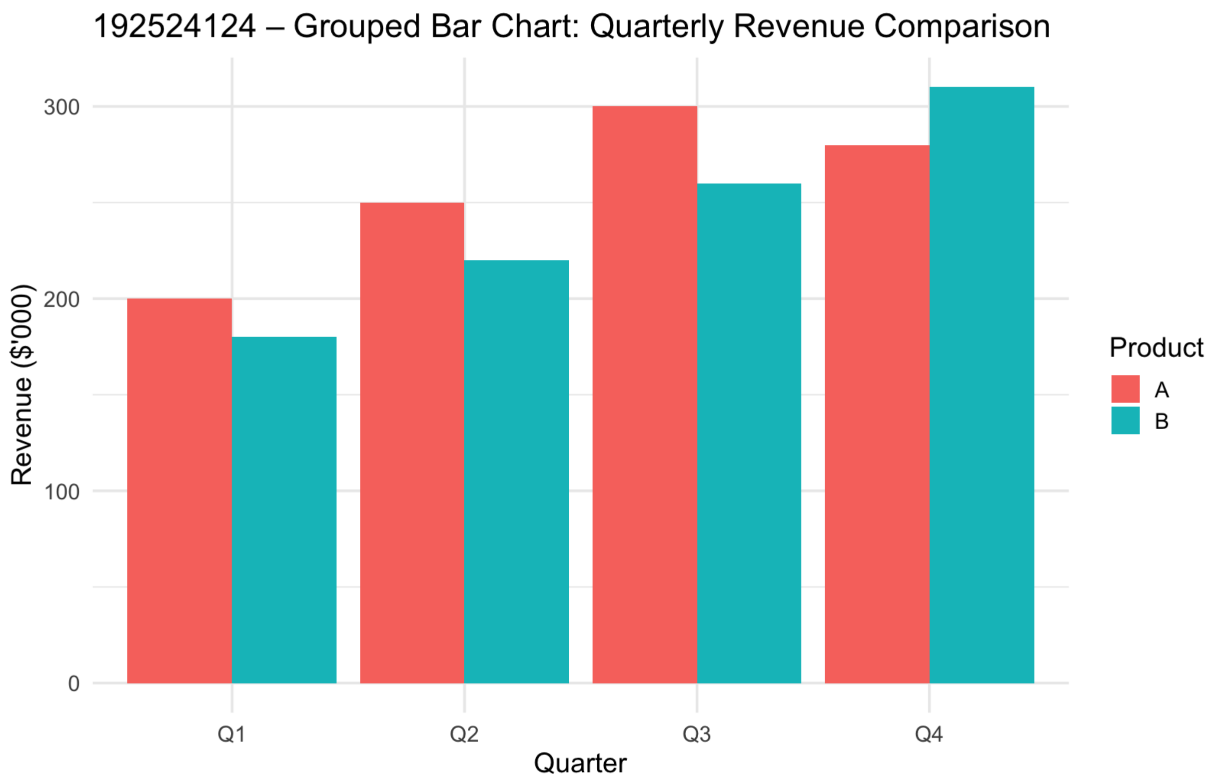

Figure 3: Grouped Bar Chart — Quarterly Revenue Comparison

Problem 4 — Stacked Bar Chart (5 Marks)

Given Data:

Year	Engineering	Sales	Support	Total
2022	40	25	15	80
2023	55	30	20	105

Task: ***Construct a stacked bar chart showing total headcount for each year, broken down by department, with a legend for each department.***

Construction Plan (Key Elements):

Chart Type	Stacked bar chart
X-Axis	Year — categorical (2022, 2023)
Y-Axis	Employees — numeric, cumulative headcount
Axis Scale	0 to 105+, in intervals of 25
Aesthetic Mapping / Legend	Colour hue = Department (Support, Sales, Engineering, stacked bottom to top)

Construction Working / Explanation:

Each year's total headcount is the sum of its three departments: 2022 = 40 + 25 + 15 = 80 employees; 2023 = 55 + 30 + 20 = 105 employees. These totals define the overall height of each bar.

Within each bar, the departments are stacked in a fixed, consistent order (Support at the base, then Sales, then Engineering at the top) so that segment boundaries can be read as cumulative totals:

- 2022 — Support: 0 to 15; Sales: 15 to 40 (15+25); Engineering: 40 to 80 (40+40).
- 2023 — Support: 0 to 20; Sales: 20 to 50 (20+30); Engineering: 50 to 105 (50+55).

The value axis is scaled from 0 to just above 105 (the larger of the two totals) in intervals of 25, and a shared legend maps each colour to its department consistently across both bars, allowing the viewer to compare both the overall headcount growth (80 → 105) and the change within each individual department between the two years.

Constructed Chart:

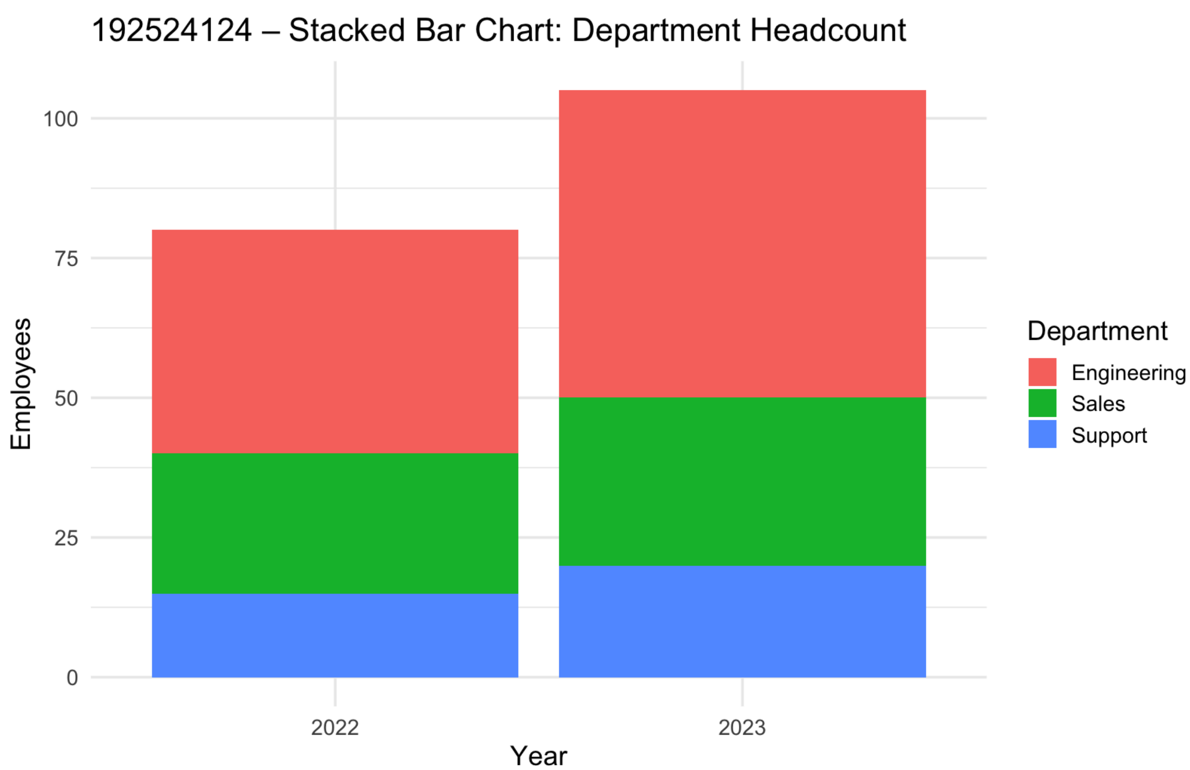

Figure 4: Stacked Bar Chart — Department Headcount by Year

Problem 5 — Line Chart — Single Series (5 Marks)

Given Data:

Day	Temperature (°C)
Mon	22
Tue	24
Wed	23
Thu	27
Fri	29
Sat	26
Sun	25

Task: ***Construct a line chart plotting temperature across the week. Mark and label each individual data point on the line.***

Construction Plan (Key Elements):

Chart Type	Line chart, single series
X-Axis	Day — categorical, sequential (Mon → Sun)
Y-Axis	Temperature (°C) — numeric, quantitative
Axis Scale	22 to 29 (data range), gridlines at intervals of 2°C
Data Labels	Each of the 7 points marked with a dot and labeled with its exact °C value

Construction Working / Explanation:

The seven daily readings (Mon = 22, Tue = 24, Wed = 23, Thu = 27, Fri = 29, Sat = 26, Sun = 25) are plotted as individual points along the day axis, in day-of-week order, and connected by straight line segments. This ordering is important: because a line chart's connecting segments imply a continuous progression, the days must be placed in their natural weekly sequence (Mon → Sun) rather than any other order, or the slope of the line would misrepresent the day-to-day trend.

The temperature (Y) axis is scaled to comfortably cover the data range of 22°C to 29°C, with labeled gridlines at intervals of 2°C, so that small day-to-day changes (as little as 1°C, e.g., Tue to Wed) remain visually distinguishable rather than being compressed by an overly wide scale.

Each data point is marked with a solid marker and its exact value is labeled directly beside it, satisfying the task's requirement to identify every individual reading rather than relying on the viewer to estimate values from the line's position against the axis alone. The overall shape of the line — a sharp dip at Mon (22°C) followed by a rise to a peak at Fri (29°C) and a gradual decline into the weekend — is preserved exactly as given in the dataset.

Constructed Chart:

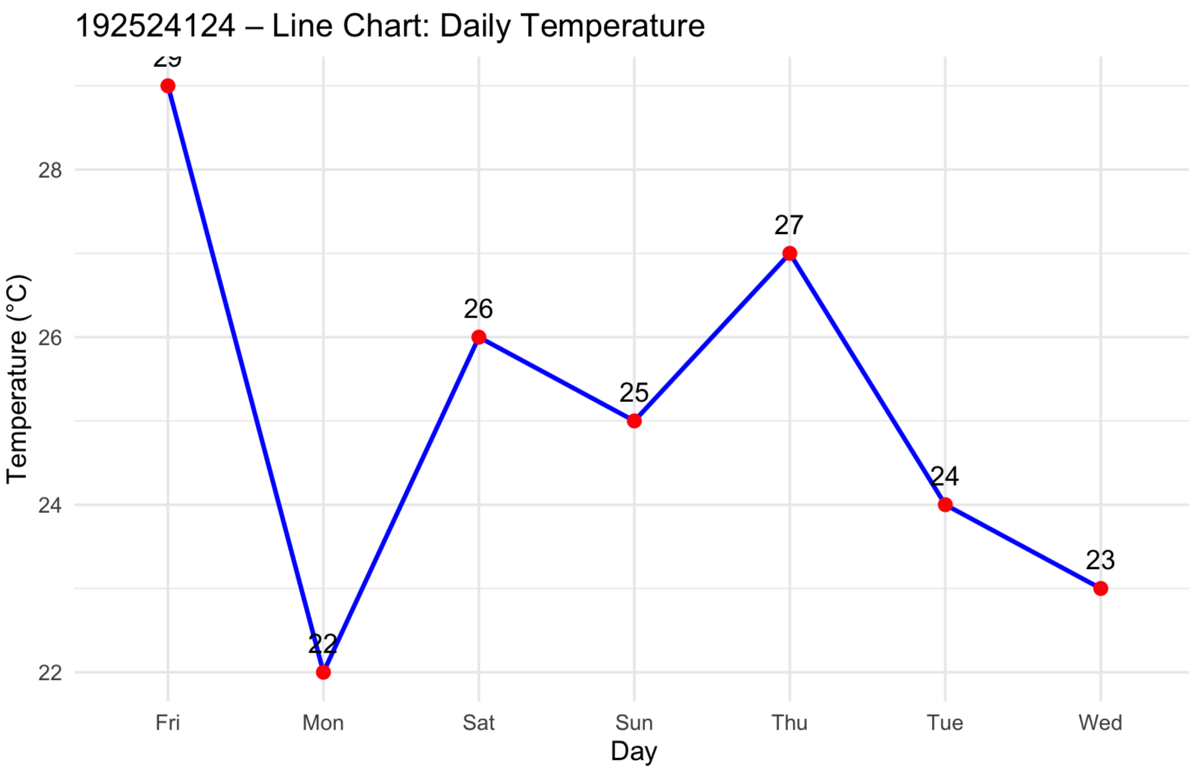

Figure 5: Line Chart — Daily Temperature for the Week